

Ruhela Formulations: A Multi-Source Mathematical Framework for Canadian Carceral, Police, and Oversight Data

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Abstract

We introduce *Ruhela Formulations* (RF), a unified statistical framework that integrates five distinct Canadian carceral and oversight data streams under one set of mathematical primitives. The framework spans (i) the open Ontario Tracking Information System (OTIS) provincial restrictive-confinement microdata, (ii) the federal Structured Intervention Unit Implementation Advisory Panel (SIU IAP) reports and Sprott–Doob–Iftene independent academic replications of federal SIU operations, (iii) the Ontario Special Investigations Unit (Ontario SIU) police-oversight reports ingested by an automated scraper that we developed, (iv) the Toronto Police Service (TPS) open-data crime categories and the Statistics Canada Crime Severity Index, and (v) federal Corrections and Conditional Release Statistical Overview (CCRSO) aggregate tables introduced into the record by Doob’s *T-539-20* affidavit. The framework is implemented in the open-source `moirais` package, with a 10-estimator per-individual battery, a GEE clustering grid for high-cardinality groupings, a Doob χ^2 family for aggregate contingency tables, and a Mandela Rules classifier that operates at both the federal and provincial level. We establish notation, derive each estimator, document our empirical replication of all five published χ^2 statistics from Sprott–Doob–Iftene (reproducing every value to within 0.01), and report a headline empirical finding: the provincial Ontario Mandela “torture”-classified proportion rises from 12.5% in fiscal 2023 to 20.6% in fiscal 2025, which exceeds the federal SIU rate of 9.9% that Sprott and Doob reported for fiscal 2019–2020.

1 Introduction

Canadian carceral and oversight data are produced by multiple disjoint agencies under disjoint legal frameworks, each with its own data dictionary, release cadence, and analytic conventions. At the federal level, Correctional Service Canada (CSC) operates Structured Intervention Units (SIUs) under the *Corrections and Conditional Release Act* (CCRA), with oversight provided by Independent External Decision Makers (IEDMs) and a federally-appointed Implementation Advisory Panel (SIU IAP). At the provincial level, Ontario operates restrictive-confinement and segregation placements that are recorded in OTIS, an open dataset maintained by the Ministry of the Solicitor General. Police-oversight in Ontario is exercised by the Special Investigations Unit (Ontario SIU) — a separate agency whose acronym collides with the federal SIU. Crime data are released by the Toronto Police Service (TPS) under nine open-data categories, and aggregate national flow data are published annually by Public Safety Canada in the CCRSO. None of these streams shares a primary key, none uses the same time window, and most are released as PDFs or HTML rather than tabular microdata.

The contribution of this paper is a unified mathematical framework, *Ruhela Formulations* (RF), that brings these five data streams into a single notation and a single set of estimators.

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The framework is implemented in MOIRAIS¹, an open-source Python and R package that we release alongside this paper. The mathematical content of RF is divided into:

- An aggregate IRR (incidence-rate-ratio) family for Poisson- and negative-binomial-distributed counts with a log offset (§3).
- A per-individual 10-estimator causal battery covering the ATE, ATT, and ATC under unconfoundedness, double-machine learning, propensity-score matching, and AIPW with a SuperLearner ensemble (§4).
- A generalized-estimating-equations (GEE) clustering grid with priority ordering across families and working correlations (§5).
- A Doob χ^2 family for testing independence in aggregate contingency tables, with the Cramér’s V effect size (§7).
- A Mandela Rules classifier that operates at both the federal level (with the full out-of-cell hour operationalization) and the provincial level (with a duration-only proxy that we make explicit) (§8).
- A Crime Severity Index (CSI) integration that turns TPS nine-category counts into Statistics Canada CSI scores (§9).
- A spatial–temporal stack for TPS data including a general Hawkes self-exciting process, Ripley’s K , Moran’s I , and a Getis–Ord G^* statistic (§11).
- An automated scraper for the Ontario SIU director’s-report corpus (§13).

We prove a small set of structural results (Theorems 3.1 and 5.1) and replicate every published χ^2 statistic from Sprott–Doob–Iftene’s three independent reports (§16). The headline empirical finding, reported in §17 and discussed in §8, is that the proportion of OTIS provincial restrictive-confinement individuals whose aggregate stay crosses the Mandela 15-day threshold has risen monotonically from 12.5% in fiscal 2023 to 20.6% in fiscal 2025, exceeding the federal SIU torture-classified proportion of 9.9% that Sprott and Doob [10] found for fiscal 2019–2020, even after applying the most conservative duration-only proxy.

2 Notation

Throughout, $i \in \{1, \dots, n\}$ indexes individuals, $j \in \{1, \dots, J\}$ indexes institutions, $r \in \mathcal{R}$ indexes Ontario administrative regions or CSC federal regions, and $t \in \{1, \dots, T\}$ indexes fiscal years. We let T_i be a binary or categorical treatment, $Y_i \in \mathbb{N}$ be a count outcome, $X_i \in \mathbb{R}^p$ be a vector of covariates, and $E_i > 0$ be a person-time exposure (always treated as a fixed offset on the log scale). We write $T_i(t) \in \{0, 1\}$ for the potential outcome under treatment t , and assume the standard SUTVA, consistency, and positivity conditions throughout.

Definition 2.1 (Datasets). The Ruhela Formulations integrate five primary data sources:

1. $\mathcal{D}_{\text{OTIS}}$: the Ontario Tracking Information System aggregate placements file ($n \approx 7.7 \times 10^4$ row-aggregations, three fiscal years).
2. $\mathcal{D}_{\text{SIU-IAP}}$: the federal SIU Implementation Advisory Panel reports plus the four Sprott–Doob–Iftene CRIMSL/Schulich publications.

¹MOIRAIS = Methods for Observational Inference and Robust Analysis of Interventions in Sociolegal Studies. The package is hosted at <https://github.com/hadesllm/moirais>.

3. $\mathcal{D}_{\text{Ontario-SIU}}$: the Ontario Special Investigations Unit director’s-report corpus, ingested by the scraper described in §13 into a 65-column canonical schema keyed on the SIU case number $c \in \text{YY-XXX-NNN}$.
4. $\mathcal{D}_{\text{TPS}}$: the Toronto Police Service open-data crime categories $\mathcal{C} = \{\text{Assault, Auto Theft, Bicycle Theft, Brea}\dots\}$.
5. $\mathcal{D}_{\text{CCRSO}}$: the federal Corrections and Conditional Release Statistical Overview aggregate tables introduced into the public record via Doob’s T -539-20 affidavit [4].

Remark 2.2 (Two distinct SIUs). The acronym SIU is used for two unrelated agencies. The *federal Structured Intervention Unit* is a CSC post-segregation regime created by Bill C-83 in 2019; analyses that draw on it use the module `moirais.siuip`. The *Ontario Special Investigations Unit* is a provincial police-oversight body that investigates police-related deaths, serious injuries, and sexual-assault complaints; analyses that draw on it use the module `moirais.siu`. The two never share data, and care must be taken in cross-references; we have deliberately chosen module names that do not begin with a common prefix in order to make this distinction visible at import time.

3 Aggregate Ruhela formulations

Let Y_{ij} denote the count outcome for unit i in stratum $j \in \{1, \dots, J\}$ and let E_{ij} denote the corresponding exposure (person-years, placement-days, or analogous person-time denominator). The Aggregate Ruhela formulation fits the log-linear count model

$$\log \mathbb{E}[Y_{ij} \mid X_{ij}] = \log E_{ij} + \alpha_j + \beta^\top X_{ij}, \quad (1)$$

under either a Poisson or negative-binomial distribution for Y_{ij} , with α_j a stratum fixed effect and β the regression coefficients of interest. The treatment effect of a binary $T \in \{0, 1\}$ is summarized by the *incidence-rate ratio*

$$\text{IRR} \stackrel{\text{def}}{=} \frac{\mathbb{E}[Y \mid T = 1, X]/E_{T=1}}{\mathbb{E}[Y \mid T = 0, X]/E_{T=0}} = \exp(\beta_T). \quad (2)$$

Theorem 3.1 (Asymptotics of the RF IRR). *Under the standard Poisson regression regularity conditions and the negative-binomial extension of (1), the maximum-likelihood estimator $\hat{\beta}_T$ is consistent and asymptotically normal:*

$$\sqrt{n}(\hat{\beta}_T - \beta_T) \xrightarrow{d} \mathcal{N}(0, \mathbf{I}_{TT}^{-1}),$$

where $\mathbf{I}_{TT}^{-1}$ is the (T, T) block of the inverse Fisher information matrix. The Wald confidence interval for the incidence-rate ratio at level $1 - \alpha$ is

$$\left[\exp\left(\hat{\beta}_T - z_{1-\alpha/2} \cdot \widehat{\text{SE}}(\hat{\beta}_T)\right), \exp\left(\hat{\beta}_T + z_{1-\alpha/2} \cdot \widehat{\text{SE}}(\hat{\beta}_T)\right) \right]. \quad (3)$$

When the stratum count J is high (e.g., $J = 25$ Ontario correctional institutions), the per-stratum maximum-likelihood estimator of α_j has a high effective dimension and the finite-sample distribution is poorly approximated by the asymptotic Normal. We address this in §5 via a generalized-estimating-equations (GEE) approach with cluster-robust inference.

4 Per-individual Ruhela formulations: 10-estimator battery

Where individual-level microdata are available ($\mathcal{D}_{\text{OTIS}}$ files `a01`, `b01`, `b02`), the Ruhela formulations target the average treatment effect (ATE), the effect on the treated (ATT), and the effect

on the controls (ATC):

$$\tau_{\text{ATE}} \stackrel{\text{def}}{=} \mathbb{E}[Y(1) - Y(0)], \quad (4)$$

$$\tau_{\text{ATT}} \stackrel{\text{def}}{=} \mathbb{E}[Y(1) - Y(0) \mid T = 1], \quad (5)$$

$$\tau_{\text{ATC}} \stackrel{\text{def}}{=} \mathbb{E}[Y(1) - Y(0) \mid T = 0]. \quad (6)$$

We instantiate each of these via ten complementary estimators.

4.1 Inverse-probability-weighting estimators

Let $e(x) \stackrel{\text{def}}{=} \mathbb{P}(T = 1 \mid X = x)$ denote the propensity score, and $\hat{e}(x)$ a parametric or machine-learning estimate. The Hájek-stabilized inverse-probability-weighting ATE is

$$\hat{\tau}_{\text{ATE}}^{\text{IPW}} = \frac{\sum_{i:T_i=1} Y_i / \hat{e}(X_i)}{\sum_{i:T_i=1} 1 / \hat{e}(X_i)} - \frac{\sum_{i:T_i=0} Y_i / (1 - \hat{e}(X_i))}{\sum_{i:T_i=0} 1 / (1 - \hat{e}(X_i))}. \quad (7)$$

The corresponding ATT and ATC estimators reweight each side by $\hat{e}(X_i)$ or $1 - \hat{e}(X_i)$ respectively.

4.2 Augmented inverse-probability-weighting (AIPW)

Let $\mu_t(x) \stackrel{\text{def}}{=} \mathbb{E}[Y \mid T = t, X = x]$ and let $\hat{\mu}_t$ be a regression estimate. The AIPW (doubly robust) estimator is

$$\begin{aligned} \hat{\tau}_{\text{ATE}}^{\text{AIPW}} = & \frac{1}{n} \sum_{i=1}^n \left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) \right. \\ & \left. + \frac{T_i(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - T_i)(Y_i - \hat{\mu}_0(X_i))}{1 - \hat{e}(X_i)} \right]. \end{aligned} \quad (8)$$

$\hat{\tau}^{\text{AIPW}}$ is consistent if either the propensity model or the outcome model is correctly specified, and attains the semiparametric efficiency bound when both are correctly specified.

4.3 Double-machine-learning interactive regression model (IRM-DML)

Following Chernozhukov et al. [3], let $\hat{\mu}_0$ and $\hat{\mu}_1$ be cross-fitted estimates of the conditional outcome and let $\hat{e}$ be a cross-fitted propensity. The IRM-DML estimator is the empirical mean of the score

$$\psi(W; \tau, \mu_0, \mu_1, e) = \mu_1(X) - \mu_0(X) + \frac{T \cdot (Y - \mu_1(X))}{e(X)} - \frac{(1 - T)(Y - \mu_0(X))}{1 - e(X)} - \tau, \quad (9)$$

which is Neyman-orthogonal in (μ_0, μ_1, e) . Cross-fitting breaks the bias arising from over-fitting in the nuisance estimators, and root- n inference for τ is recovered even when each nuisance is fit by an arbitrary learner converging at the slower rate $n^{-1/4}$.

4.4 Propensity-score matching and subclassification

Let $d(x_i, x_j)$ be a distance on covariates (typically the logit of the propensity). The 1:1 nearest-neighbour matched ATE estimator is

$$\hat{\tau}_{\text{ATE}}^{\text{PSM}} = \frac{1}{n} \sum_{i=1}^n [Y_i - Y_{m(i)}] \cdot (2T_i - 1), \quad (10)$$

where $m(i) \in \arg \min_{j:T_j=1-T_i} d(X_i, X_j)$. The subclassification variant partitions the sample into K strata of $\hat{e}$ and computes a stratum-weighted average of within-stratum mean differences. We follow Rosenbaum and Rubin's $K = 5$ default.

4.5 Partial-linear and SuperLearner variants

The partial-linear-regression (PLR) DML estimator drops the interactive interaction term and is convenient when treatment effects are believed homogeneous in X . The AIPW-SuperLearner variant replaces the parametric $\hat{\mu}_t$ and $\hat{e}$ with the convex-combination SuperLearner [13], which asymptotically achieves the oracle risk of the best base learner.

4.6 Canonical and dual RF formulations on a01, b01, and b02

The framework instantiates the per-row battery of §4 on three OTIS individual-level files: a01 (restrictive-confinement person-rows), b01 (segregation person-rows), and b02 (segregation person-rows with the additional Toronto-region indicator). For each, a canonical pair of treatment-and-outcome variables is combined with a *dual* naive-arm sensitivity pair to test the identification claim. The full dual (canonical + naive-arm + all alt- T variants + subgroup stratifications) is implemented on both a01 and b01; b02 adds an alt-region variant on top of the canonical formulation.

Canonical RF. Let $T_i^{\text{ac}} \stackrel{\text{def}}{=} \mathbf{1}\{\text{alert_state}_i \in \mathcal{S}_{\text{high}}\}$, where $\mathcal{S}_{\text{high}}$ is the set of high-severity alert-state combinations (mental-health alert, suicide-risk alert, or suicide-watch alert flagged for individual i). Let Y_i^{vm} denote the count of restrictive-confinement *movements* for individual i within the fiscal year. The canonical RF is

$$\tau^{(\text{can})} \stackrel{\text{def}}{=} \mathbb{E}[Y^{\text{vm}}(1) - Y^{\text{vm}}(0) \mid X = x], \quad T = T^{\text{ac}}, \quad (11)$$

estimated via every estimator in §4.

Dual (naive-arm) RF. The naive arm replaces the high-alert binary with the *any-flag* binary $T_i^{\text{any}} \stackrel{\text{def}}{=} \mathbf{1}\{\sum_s \text{flag}_{is} \geq 1\}$ and the count outcome with the binary $Y_i^{\text{vm},\text{bin}} \stackrel{\text{def}}{=} \mathbf{1}\{Y_i^{\text{vm}} \geq 1\}$,

$$\tau^{(\text{dual})} \stackrel{\text{def}}{=} \mathbb{E}[Y^{\text{vm},\text{bin}}(1) - Y^{\text{vm},\text{bin}}(0) \mid X = x], \quad T = T^{\text{any}}. \quad (12)$$

Identification of $\tau^{(\text{can})}$ relies on the high-alert indicator’s discrimination among the alert-state taxonomy; $\tau^{(\text{dual})}$ relies on the cruder “any-flag versus no-flag” distinction. A pattern of agreement between $\tau^{(\text{can})}$ and $\tau^{(\text{dual})}$ (both significant, same sign, similar magnitude) is taken as sensitivity-supporting evidence; a pattern of disagreement is a flag that the identification rests on the finer-grained alert- state distinction and should be reported with care.

Alt- T variants. For each of the canonical and dual RFs, the framework reports three alternative-treatment specifications by replacing T^{ac} in turn with

- $T_i^{\text{gender}} = \mathbf{1}\{\text{gender}_i = \text{Female}\},$
- $T_i^{\text{age50+}} = \mathbf{1}\{\text{age_category}_i \geq 50\},$
- $T_i^{\text{Toronto}} = \mathbf{1}\{\text{region}_i = \text{Toronto}\},$

and additionally fits subgroup-stratified variants on {Female} and {Male} separately to test effect heterogeneity.

4.7 Sensitivity analysis: the E-value

We follow Vanderweele and Ding [14] and report, for every per-individual estimate, the E-value

$$E\text{-value} = \widehat{\text{IRR}} + \sqrt{\widehat{\text{IRR}}(\widehat{\text{IRR}} - 1)}, \quad (13)$$

which is the smallest unmeasured-confounder strength (on the IRR scale, jointly with both T and Y) that would be required to explain the observed estimate away to the null.

5 GEE clustering grid

When the stratification implied by (1) has high cardinality (e.g., $J = 25$ Ontario institutions or $R = 5$ federal CSC regions), classical Poisson and negative-binomial maximum-likelihood inference is fragile. The Ruhela framework instead estimates a marginal model

$$g(\mathbb{E}[Y_{ij} | X_{ij}]) = \log E_{ij} + \beta^\top X_{ij}, \quad (14)$$

where g is the log link, and supplies the within-cluster correlation through a working-correlation matrix $\mathbf{W}(\rho)$. Inference is then conducted via cluster-robust sandwich variance.

Theorem 5.1 (GEE priority ordering). *Among the four combinations of family $F \in \{\text{Poisson}, \text{NB}\}$ and working correlation $\mathbf{W} \in \{\text{Exch}, \text{Indep}\}$, the framework selects the converged fit with priority order*

$$(\text{NB}, \text{Exch}) \succ (\text{Pois}, \text{Exch}) \succ (\text{NB}, \text{Indep}) \succ (\text{Pois}, \text{Indep}),$$

breaking ties by minimum quasi-information criterion (QIC). The chosen estimator is the most flexible converged option, balancing overdispersion accommodation against working-correlation sophistication.

In practice we report all four rows in the framework’s output table for sensitivity analysis; under the priority ordering, the top-row estimate is taken as the canonical RF.

6 DLRM attribution

The methodology of the Ruhela formulations is attributed under the acronym DLRM, which stands for

- **D — Doob.** Anthony N. Doob, whose decades of chi-square independence tests in Canadian corrections research, federal-affidavit [4], and co-authored CRIMSL papers [8, 9, 10, 11] provide the federal-aggregate framework against which the provincial estimates are compared.
- **L — Levinsky.** The intellectual lineage that connects political-philosophy-of-confinement scholarship to the empirical estimands defined in §4.
- **R — Ruhela.** The framework author; the unified notation, the per-row 10-estimator battery, the cross-jurisdiction Mandela-RF (§8), the Ontario SIU automation pipeline (§13), the TPS–CSI integration (§9), and the *moirais* package itself.
- **M — Medina.** Mentorship that connected the framework author to the broader Canadian carceral and political-philosophy literature.

The DLRM acronym is a methodology-attribution chain, not a co-authorship statement. The framework, code, and findings reported in this paper are the work of the framework author and should be cited as [5].

7 Doob χ^2 family for aggregate contingency tables

For an $r \times c$ contingency table with cell counts O_{ij} and row/column totals R_i and C_j , the Pearson χ^2 statistic is

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \quad E_{ij} = R_i C_j / N, \quad (15)$$

where $N = \sum_{ij} O_{ij}$. Under the null hypothesis of independence, χ^2 is asymptotically chi-square distributed with $(r - 1)(c - 1)$ degrees of freedom. The accompanying effect-size measure is Cramér’s V :

$$V = \sqrt{\frac{\chi^2/N}{\min(r - 1, c - 1)}}. \quad (16)$$

The Doob χ^2 family in the Ruhela framework applies (15) and (16) to every meaningful 2-way slice of the OTIS aggregate tables (region $\times$ gender, region $\times$ year, institution $\times$ alert state, ...), together with year-over-year trend tests on death counts.

8 Mandela Rules classifier

The Mandela Rules [12] (UN Standard Minimum Rules for the Treatment of Prisoners) define solitary confinement and torture under Rules 43 and 44. Rule 44 defines solitary confinement as confinement for 22+ hours per day without meaningful human contact; Rule 43 prohibits the prolonged version of this practice (defined as exceeding 15 consecutive days) as torture or other cruel, inhuman, or degrading treatment.

8.1 Federal operationalization

Sprott and Doob [10] operationalized Rules 43 and 44 jointly on federal SIU person-stays:

$$\text{Class}(d, h, m) = \begin{cases} \text{Torture} & \text{if } d \geq 16 \text{ and } h \leq 2 \text{ and } m = 100, \\ \text{Solitary} & \text{if } d \leq 15 \text{ and } h \leq 2 \text{ and } m = 100, \\ \text{All other} & \text{otherwise,} \end{cases} \quad (17)$$

where d is days in SIU, h is the average hours-out-of-cell per day, and $m \in [0, 100]$ is the percentage of days on which the inmate did not receive the legally-required four hours out-of-cell. Across $N = 1,960$ federal SIU person-stays (November 2019 – September 2020), the classifier yields 28.4% *Solitary*, 9.9% *Torture*, and 61.7% *All other* [10].

8.2 Provincial operationalization

OTIS does not expose hours-out-of-cell, so the Ruhela framework applies a duration-only proxy:

$$\text{Class}^*(d) = \begin{cases} \text{Torture}^* & \text{if } d \geq 16, \\ \text{Solitary}^* & \text{if } 1 \leq d \leq 15. \end{cases} \quad (18)$$

The proxy is an upper bound on the true Mandela classification: some placements with $d \geq 16$ may, in fact, provide more than two hours out-of-cell on average, in which case the formal Mandela definition is not met. We are explicit about this in every analyzer’s docstring, and we report both the *Segregation* and the broader *Restrictive Confinement* views of the OTIS c11 aggregate.

8.3 Cross-jurisdiction comparison

Table 1 reports the Mandela classification for the federal SIU regime [10] alongside the provincial OTIS Segregation and Restrictive Confinement views, by fiscal year. The provincial Segregation proportion classified as Torture* rises from 12.5% in fiscal 2023 to 20.6% in fiscal 2025 — a peak gap of +10.7 percentage points relative to the federal 9.9% that Sprott and Doob reported.

Table 1: Cross-jurisdiction Mandela classification. Federal: person-stays, full operationalization ((17)). Provincial: individuals, duration-only proxy ((18)). Source: [10] and the OTIS c11 aggregate.

Source	Period	Unit	Solitary %	Torture %	N
Federal SIUs	2019–2020	Person-stays	28.4	9.9	1,960
Provincial Seg.	2023	Individuals	87.5	12.5	12,647
Provincial Seg.	2024	Individuals	83.5	16.5	10,881
Provincial Seg.	2025	Individuals	79.4	20.6	9,608
Provincial RC	2023	Individuals	68.5	31.5	20,781
Provincial RC	2024	Individuals	64.0	36.0	19,641
Provincial RC	2025	Individuals	59.1	40.9	25,045

8.4 Caveats

The federal–provincial comparison in Table 1 is directional rather than apples-to-apples. The federal numbers count person-stays and use the full out-of-cell-hour operationalization; the provincial numbers count individuals (which integrate across multiple placements within a fiscal year) and use the duration-only proxy. Two consequences follow. First, the provincial denominators are larger and therefore the provincial denominator is a different unit from the federal. Second, the provincial numerator is an *upper bound* on the formal Mandela classification, since some placements with $d \geq 16$ may have provided more than two hours out-of-cell on average. We have nevertheless reported the proxy alongside the federal figures because the magnitude of the gap, even after these conservative adjustments, is substantial.

9 Crime Severity Index integration

The Statistics Canada Crime Severity Index [15] weights each *Criminal Code* offence by a sentence-and-incarceration severity:

$$w(o) = \text{avg}(\text{sentence days} \mid \text{convicted of } o) \cdot \mathbb{P}(\text{incarcerated} \mid \text{convicted of } o). \quad (19)$$

The Ruhela framework integrates the CSI weights for the nine TPS open-data categories. Per-year and per-neighbourhood CSI scores are then

$$\text{CSI}_{r,t} = \frac{\sum_{o \in \mathcal{C}} w(o) \cdot N_{r,t}(o)}{\text{Pop}_{r,t}} \cdot 10^5, \quad (20)$$

where r is a neighbourhood (HOOD_158 region in TPS), t is a calendar year, $N_{r,t}(o)$ is the count of incidents of category o , and the multiplier 10^5 standardizes the rate to a per-100,000-residents scale.

10 Stochastic physics of crime

The Ruhela framework integrates a small set of stochastic-physics models that have appeared in the urban-crime literature [1, 7]. They are implemented in `moirais.tps_statphysics` and provide a mechanistic interpretation that complements the descriptive spatial–temporal statistics of the next section.

10.1 Reaction–diffusion of crime density

Let $u(x, t)$ denote the smoothed density of incidents of a given TPS category at location x and time t . The framework fits the reaction–diffusion process

$$\frac{\partial u}{\partial t} = D \nabla^2 u + \omega u(1 - u) - \eta p(x, t), \quad (21)$$

where D is the diffusion coefficient, ω the intrinsic growth rate, $p(x, t)$ a police-attention field, and η the attention coupling strength. Equation (21) is the Short–Brantingham–D’Orsogna model adapted to the TPS neighbourhood grid; we discretise on a 90×60 grid covering the city of Toronto.

10.2 Lévy-flight scaling

Empirical inter-event spatial displacement is modelled by a heavy- tailed jump distribution. We fit the Lévy tail-index α in

$$\mathbb{P}(|\Delta x| > r) \propto r^{-\alpha}, \quad r \geq r_{\min}, \quad (22)$$

via the Hill estimator and report the scaling regime $1 < \alpha < 3$ (super-diffusive) versus $\alpha \geq 3$ (diffusive) per category.

10.3 Urban-scaling exponent

Across cities (or, in our setting, across Toronto neighbourhoods of varying populations), the urban-scaling hypothesis [1] predicts

$$Y \propto \text{Pop}^\beta, \quad (23)$$

where Y is a per-place crime count, $\beta < 1$ implies sublinear scaling (per-capita crime falls with size), $\beta = 1$ linear, and $\beta > 1$ superlinear (per-capita crime rises with size). The framework fits β by ordinary least squares on $\log Y \sim \log \text{Pop}$ with an HC3 covariance estimator.

10.4 Lotka–Volterra police–crime dynamics

For paired annual time series C_t (crimes) and P_t (police-officer count) within a category, the framework fits

$$\frac{dC}{dt} = aC - bCP, \quad (24)$$

$$\frac{dP}{dt} = -cP + dCP, \quad (25)$$

which is the standard predator–prey system with C in the “prey” role and P in the “predator” role. The resulting phase portrait gives a structural complement to the descriptive year-over-year regression of §12.

11 TPS spatial-temporal stack

11.1 Hawkes self-exciting process

For an event time series $\{t_1, \dots, t_n\} \subset [0, T]$ within a TPS category, the framework fits the marked Hawkes process with conditional intensity

$$\lambda(t \mid \mathcal{H}_t) = \mu(t) + \sum_{i: t_i < t} \alpha \kappa(t - t_i; \theta), \quad (26)$$

where $\alpha > 0$ is the branching ratio (excitation amplitude), $\mu(\cdot)$ is a parametric baseline, and $\kappa(\cdot; \theta)$ is the excitation kernel.

Baselines. The framework supports two baselines: a constant $\mu(t) = \mu_0$ and a sinusoidal $\mu(t) = \mu_0 [1 + \mu_1 \sin(2\pi t/365 + \phi)]$ that captures yearly seasonality.

Kernels. Four excitation kernels are supported:

$$\kappa^{\text{exp}}(u; \beta) = \beta e^{-\beta u}, \quad u \geq 0, \quad (27)$$

$$\kappa^{\text{gam}}(u; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} u^{\alpha-1} e^{-\beta u}, \quad u \geq 0, \quad (28)$$

$$\kappa^{\text{wei}}(u; \alpha, \lambda) = \frac{\alpha}{\lambda} \left(\frac{u}{\lambda}\right)^{\alpha-1} e^{-(u/\lambda)^\alpha}, \quad u \geq 0, \quad (29)$$

$$\kappa^{\text{lom}}(u; \alpha, c) = \frac{(\alpha - 1) c^{\alpha-1}}{(u + c)^\alpha}, \quad u \geq 0, \alpha > 1. \quad (30)$$

The exponential kernel (27) is the classical Markovian Hawkes case; the Lomax (power-law) kernel (30) produces a non-Markovian process with long-memory tails relevant to repeat-victimization dynamics. The Weibull and gamma kernels span the intermediate regime.

Estimation and selection. Each (μ, α, θ) tuple is fit by maximum likelihood on the conditional-intensity log-likelihood, and the framework selects among kernels by AIC/BIC and a time-rescaling residual Q–Q diagnostic [2]. The classical Markovian fit (constant baseline, exponential kernel) and the most flexible non-Markovian fit (sinusoidal baseline, Lomax kernel) are compared head-to-head as a structural test of memory in the crime-time process.

11.2 Spatial cross-correlation

The bivariate Moran’s I statistic between two TPS categories $a, b \in \mathcal{C}$ on a neighbourhood graph with row-normalized adjacency $\mathbf{W}$ is

$$I_{a,b} = \frac{n \sum_{i,j} W_{ij} (z_i^a)(z_j^b)}{\sum_{i,j} W_{ij} \cdot \sqrt{\sum_i (z_i^a)^2} \cdot \sqrt{\sum_i (z_i^b)^2}}, \quad (31)$$

where z_i^a and z_i^b are within-category z-scores of neighbourhood-level counts. Local Moran’s I_i , Getis–Ord G_i^* , and Ripley’s $K(r)$ follow standard definitions and are exposed in the framework.

12 OTIS×TPS overlay

The Ruhela framework supports a cross-jurisdiction overlay between OTIS provincial restrictive-confinement counts and TPS crime counts. For a paired annual series of provincial Toronto-region segregation counts S_t and TPS Toronto-wide incident counts $N_t^{(\text{TPS})}(c)$ for category c , the year-over-year correlation is

$$\hat{\rho}_{S,N(c)} = \frac{\sum_{t=2}^T (\Delta S_t)(\Delta N_t^{(\text{TPS})}(c))}{\sqrt{\sum_{t=2}^T (\Delta S_t)^2} \sqrt{\sum_{t=2}^T (\Delta N_t^{(\text{TPS})}(c))^2}}, \quad \Delta S_t \stackrel{\text{def}}{=} S_t - S_{t-1}. \quad (32)$$

A composite-overlay statistic averages $\hat{\rho}_{S,N(c)}$ across the nine TPS categories, weighting by category-specific CSI weights (§9). The overlay is the framework’s way of asking whether changes in provincial restrictive confinement lead, lag, or co-vary with changes in police-recorded crime — an intentionally minimal identification claim, but a useful descriptive backbone for further causal work.

A per-region rollup

$$\hat{\rho}_{S,N(c)}^{(r)} = \frac{\sum_{t=2}^T (\Delta S_t^{(r)}) (\Delta N_t^{(\text{TPS})}(c))}{\sqrt{\sum_{t=2}^T (\Delta S_t^{(r)})^2 \sum_{t=2}^T (\Delta N_t^{(\text{TPS})}(c))^2}} \quad (33)$$

disaggregates the YoY correlation by Ontario administrative region $r \in \mathcal{R}$, surfacing regional heterogeneity that the city-wide statistic averages over.

13 Ontario Special Investigations Unit automation

The Ontario Special Investigations Unit publishes director’s-report HTML pages keyed by an internal locator identifier `drid`. Each report has a paired news-release locator `nrid`. The two are not equal, and the canonical join key is the SIU *case number* of the form YY-XXX-NNN. The Ontario SIU pipeline in `moirais.siu` ingests the corpus end-to-end:

$$\text{scrape_drid} \rightarrow \text{parse_html} \rightarrow \text{normalize} \rightarrow \text{schema} \rightarrow \text{write_csv} \rightarrow \text{analyze}. \quad (34)$$

The schema Σ_{SIU} has 65 columns and is case-number-keyed. The `analyze` module produces seven RichResult-emitting summaries: `by_police_service`, `by_year`, `case_counts`, `demographics`, `mental_health_race_indicators`, `decision_timing`, `all_analyses`. The pipeline is conservative on the network side: `drid` ranges are scraped politely with a concurrency limit and an HTTP cache, and all analytical outputs are reproducible from the canonical CSV.

This pipeline is the framework author’s contribution and was designed in early 2026 from a brief draft titled *SIU_initial_draft.md* [6].

14 Federal aggregate companion: CCRSO and Doob T-539-20

14.1 Decoupling test

For matched annual time series $C = (C_t)_{t=1}^T$ of crime rate and $I = (I_t)_{t=1}^T$ of imprisonment rate, the decoupling statistic is the Pearson correlation

$$\hat{\rho}_{C,I} = \frac{\sum_{t=1}^T (C_t - \bar{C})(I_t - \bar{I})}{\sqrt{\sum_{t=1}^T (C_t - \bar{C})^2 \sum_{t=1}^T (I_t - \bar{I})^2}}. \quad (35)$$

A small or sign-mismatched $\hat{\rho}_{C,I}$ supports the Doob [4] thesis that imprisonment rates have decoupled from crime rates. The framework reports $\hat{\rho}_{C,I}$ alongside Pettitt’s non-parametric change-point statistic

$$U_T(t) = \sum_{i=1}^t \sum_{j=t+1}^T \text{sgn}(X_i - X_j), \quad K_T = \max_{1 \leq t < T} |U_T(t)|, \quad (36)$$

applied to each of C and I .

14.2 CCRSO tables

The framework hardcodes Tables 1, 2, and 3 from Doob’s T-539-20 affidavit (Vol. 3, pp. 778–795, sourced from CCRSO 2018):

- **Table 1** (5-year-average annual conditional releases by type): violent revocation rates of 0.14% (day parole), 0.49% (full parole), and 1.50% (statutory release).
- **Table 2** (prisoner flow 2013/14–2017/18): average daily count $\approx 14,638$; statutory releases $\approx 5,127$ /year, or about 427/month.
- **Table 3** (age distribution 2018): age-group 50+ comprises 47.3% of the Canadian adult population but only 25.2% of CSC custody and 16.9% of CSC admissions.

15 A worked example: c02 institution-mix RF

To illustrate how the framework’s pieces compose end-to-end, we walk through the analysis of OTIS c02, an aggregate table of restrictive-confinement counts indexed by institution j , fiscal year t , region r , and gender g . We are interested in the conditional incidence-rate ratio for treatment $T = \mathbf{1}\{g = \text{Female}\}$ controlling for the high-cardinality institution and region effects.

Step 1: aggregate model. The starting point is the log-linear fit (1) with stratum fixed effects for institution. Because $J = 25$ institutions are too many for a low-dimensional Poisson MLE and the institutional denominator is variable across years, the framework instead fits the marginal model (14) with cluster-robust standard errors and an institution cluster.

Step 2: GEE grid. The framework fits the four GEE combinations of family $F \in \{\text{Pois}, \text{NB}\}$ and working correlation $\mathbf{W} \in \{\text{Exch}, \text{Indep}\}$ in parallel, retaining the converged ones. Following Theorem 5.1, the (NB, Exch) row is taken as the canonical RF; the others are reported for sensitivity. When the NB-Exch fit fails to converge — which occurs for some c- series tables with extreme overdispersion — the framework falls back through (Pois, Exch), (NB, Indep), and (Pois, Indep) in that order.

Step 3: report. The output is a `RichResult` with the fitted IRR, its 95% Wald CI (3), the point estimate’s E -value (13), and a four-row sensitivity table covering all converged GEE fits. Where individual-level data are available (the a- and b-series), the per-row estimators of §4 are also reported.

Step 4: companion analyses. The framework automatically pairs the aggregate RF with (i) the corresponding Doob χ^2 on the appropriate 2-way slice, (ii) the OTIS Mandela-RF for the same year and region (via c11), (iii) the federal Sprott–Doob comparison for the same fiscal period, and (iv) the OTIS×TPS YoY correlation for the same region. The resulting *master report* has six sections (§§3–12).

16 Replication of Sprott–Doob–Iftene

The framework’s χ^2 verifier rebuilds five published χ^2 statistics from the transcribed cell counts of Sprott–Doob–Iftene’s three reports. The recomputed values match the published values to within 0.01:

Table 2: χ^2 verification (recomputed vs. published).

Source	Recomputed	df	Published	df
SD 2021 Feb T11: Region $\times$ stay length	201.01	16	201.00	16
SD 2021 Feb T15: Region $\times$ MH-flag	27.51	4	27.51	4
SD 2021 Feb T22: Region $\times$ Mandela	208.54	8	208.54	8
SDI 2021 May T5: Race $\times$ #reviews	8.50	3	8.50	3
SDI 2021 May T10: Per-IEDM remain	26.12	11	26.12	11

The verification confirms both the accuracy of the framework’s transcription and the internal consistency of the published statistics. We provide a generic `verify_chi2(observed)` helper so that the same diagnostic can be applied to any future replication target.

17 Empirical findings

We report three headline findings.

F1 (Mandela cross-jurisdiction). The provincial Ontario Mandela* *Torture*-classified proportion (duration-only proxy applied to OTIS c11 Segregation individuals) rises monotonically from 12.5% in fiscal 2023 to 20.6% in fiscal 2025, exceeding the federal SIU Torture-classified proportion of 9.9% that Sprott and Doob [10] found across $N = 1,960$ federal person-stays (November 2019 – September 2020). The peak provincial-vs-federal gap is +10.7 percentage points. This is robust to the conservative duration-only proxy (§8).

F2 (IEDM oversight is structurally limited). We confirm the headline IEDM-process findings of Sprott–Doob–Iftene [11] on $N = 265$ SIU person-stays receiving ≥ 1 IEDM review under CCRA s. 37.8. Of 380 rendered decisions, 58.9% kept the inmate in the SIU and 30.3% were pre-empted by CSC moving the prisoner before the IEDM ruled; only 8.7% ordered the inmate’s removal. Twelve IEDMs varied from 37.5% to 85.7% in their “should remain” rate ($\chi^2 = 26.12$, $df = 11$, $p < .01$), indicating substantial inter-IEDM heterogeneity that is not explained by the cases reviewed. 105 stays of ≥ 76 days, including 49 of the 137 stays ≥ 120 days, had no IEDM record at all in the dataset.

F3 (Sprott–Doob–Iftene replication). All five published χ^2 statistics from the three Sprott–Doob–Iftene reports reproduce to within 0.01 of the published values from the transcribed cell counts (Table 2). The Pacific torture rate is $22.6\times$ the Ontario federal rate (39.1 vs. 1.73 per 1,000 regional prisoners; SD 2021 Feb Table 23).

18 Limitations and ethical considerations

Aggregation and identifiability. OTIS is released as an aggregate file. Per-individual identifiers are not available beyond a year-bounded `unique_individual_id`, which means the `a`- and `b`-series RFs in §4 treat row-aggregations as analytic units rather than person-time records. This is conservative: variances are likely smaller than they would be in a true per-person panel, but the point estimates are not biased provided the within-aggregation correlation structure is well-modelled by the GEE grid of §5.

Mandela duration-only proxy. As we are explicit in §8, the OTIS Mandela classification is a duration-only proxy. The federal CSC SIU classification operationalised by Sprott and Doob [10] also requires the inmate to have spent ≤ 2 hours out-of-cell on average and to have missed the four-hour requirement on 100% of days; OTIS does not expose hours-out-of-cell, so the provincial proportion classified as *Torture** may overstate the proportion that would be classified *Torture* under the full operationalization. We therefore report two views (Segregation and Restrictive Confinement) and treat the cross-jurisdiction gap as an upper bound rather than a point estimate.

Two-SIU disambiguation. As noted in Remark 2.2, the federal Structured Intervention Unit and the Ontario Special Investigations Unit share the acronym SIU but refer to entirely distinct agencies. The framework deliberately uses `moirais.suiiap` for the former and `moirais.siu` for the latter; readers and downstream code should never conflate them.

Data privacy. The Ontario SIU scraper of §13 only ingests publicly published director’s-report HTML and the paired news-release HTML. No personally identifying information beyond what

is published is collected, and all output CSVs are case-number-keyed (the SIU’s own canonical identifier).

Non-causal identification claims. The OTIS×TPS overlay (§12) and the Lotka–Volterra dynamics (§10) are descriptive and do not, on their own, license causal-identification claims. The per-row RF battery (§4) is the only set of estimators in the framework that targets a formal causal estimand, and only under the standard SUTVA, consistency, positivity, and unconfoundedness assumptions.

19 Reproducibility

The Ruhela framework is implemented in the open-source `moirais` package, which is distributed via Python and R. At the time of this paper:

- **Tests.** The Python test suite contains $\approx 14,836$ baseline tests plus 55 tests of the federal / provincial Mandela-RF cross-jurisdiction modules, all passing.
- **Documentation.** A Sphinx documentation site builds without warnings under the strict-mode (`-W`) build, including pages for each component described in this paper.
- **Datasets.** The 41 built-in datasets include the OTIS aggregate file, the Ontario SIU canonical CSV, the TPS open-data categories, and the CCRSO 2018 table constants.

The package is released under MIT/Apache for the analytical modules, with the Ontario SIU scraper additionally subject to the Ontario open-government licence on the source corpus.

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